Software Requirements Specification (SRS)

Project Title: Flight Delay Analysis

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1. Introduction:

• 1.1 Purpose:

This project looks at flight data to find useful patterns in delays, helping people make better decisions and improve planning.

• 1.2 Scope:

• Data Ingestion (Read):

Loading large-scale flight datasets using optimized Python libraries.

• Data Cleaning (Clean):

Detecting outliers and performing mean/median imputation for missing values.

• Exploratory Data Analysis (Analyze):

Identifying patterns and correlations across airlines and schedules.

• **Visualization:**

Generating high-fidelity charts, including Violin Plots and interactive maps.

• Reporting:

Delivering actionable insights and recommendations to stakeholders.

• 1.3 Overview:

This project focuses on analyzing flight delay data in a clear and simple way. The results will help understand delay behavior and support better decision-making.

2. General Description

• 2.1 Users:

•The system can be useful for:

- ✓ Airline managers
- ✓ Data analysts

- ✓ Airport operation teams
- ✓ Non-technical users who want clear results

• **2.2 Objectives:**

• The project aims to:

- ❖ Identify the main causes of flight delays
- ❖ Find which airlines have higher delays
- ❖ Analyze delay patterns during the week
- ❖ Understand relationships between departure and arrival delays
- ❖ Support better planning and scheduling decisions

3. Data Requirements

- ➢ Dataset in **CSV** format

- ➢ Flight information **such as**:
  - Airline
  - Departure Delay
  - Arrival Delay
  - Day of Week
  - Delay Causes
  - Flight Status

- ➢ Data should be **complete** and **valid** for accurate analysis

4. Functional Requirements

• 4.1 The system should:

- Read the flight dataset
- Check missing values (null values)
- Fill or handle missing data
- Explore the data using functions like:
  - head()
  - tail()
  - unique()
  - value_counts()
- Perform data analysis such as:
  - Average delays
  - Delay distribution
  - Delay causes

    - Delay trends by day
- Generate visualizations such as:
    - Bar chart
    - Line chart
    - Scatter plot
    - Pie chart
    - Histogram
    - Violin plot
    - Hierarchical
    - Dount chart
- Output useful insights and findings

5. Non-Functional Requirements

- ❖ Results should be accurate
- ❖ Charts should be clear and easy to read
- ❖ Output should be simple to understand
- ❖ Code should be organized and readable
- ❖ Analysis should run smoothly without errors

6. Tools & Technologies

• 6.1 Programming Language: Python

• 6.2 Libraries:

    - NumPy
    - Pandas
    - Matplotlib
    - Plotly (for advanced charts)
    - Seaborn

• 6.3 Platform: Jupyter Notebook

7. Expected Output

• 7.1 The project is expected to produce:

- A clean and organized dataset
- Charts and graphs for data visualization
- A simple written report explaining the analysis
- Key insights about flight delays
- Recommendations to reduce delays and improve scheduling

Figure.1 output prototype

8. Constraints

• 8.1 some limitations of this project include:

- Missing or incomplete data in the dataset
- Limited time for full analysis
- Using basic tools only (Python and simple libraries)
- Results depend on the quality of available data

• 8.2 Data Quality: Managing high percentages of missing data in specific flight metrics.

• 8.3 Resources: Hardware limitations when processing millions of data rows.

• 8.4 Schedule: Project completion within the assigned academic or business timeline.

9. Schedule

• **9.1 Project management time line include as shown below:**

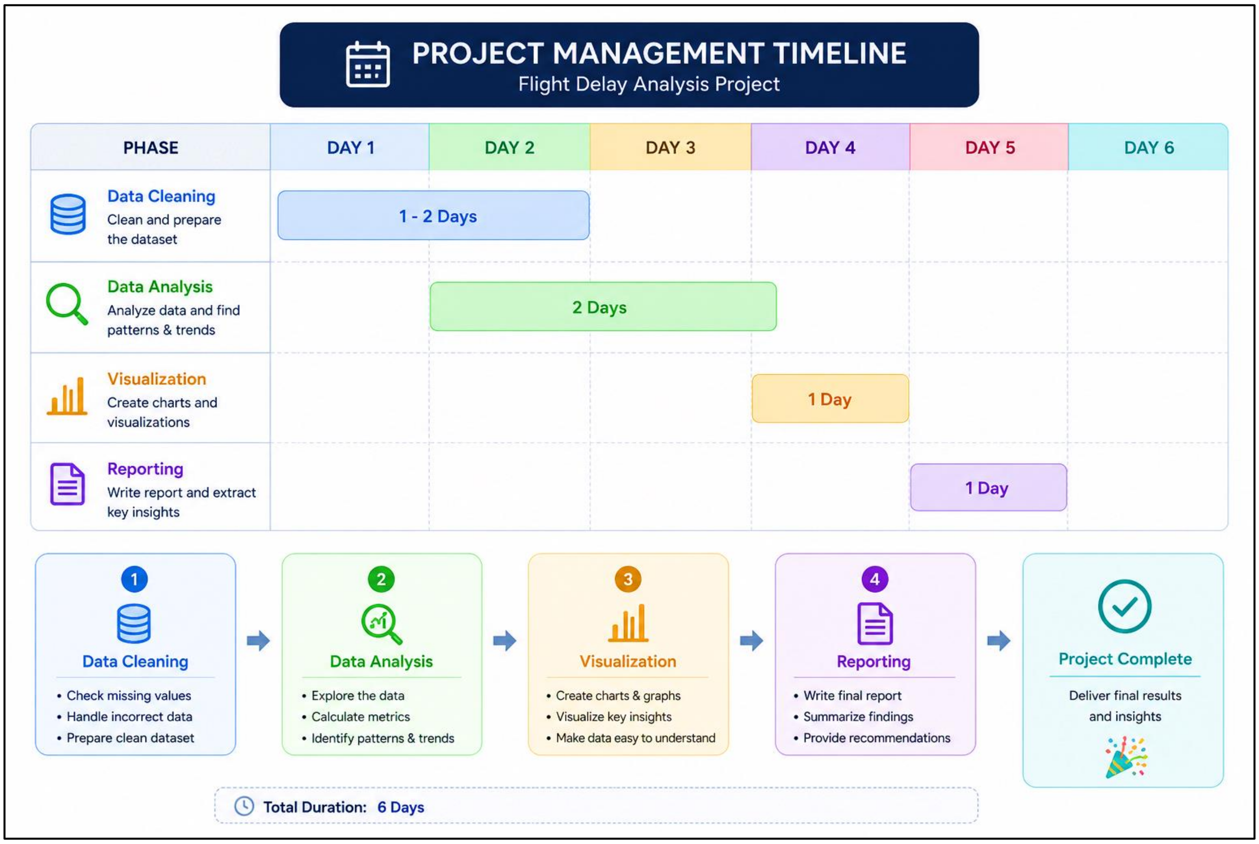

Figure.2 Project management time line

10. Financial

• **10.1.1** The project requires limited resources, mainly related to development effort and computing tools.

• **10.1.2** The main investment is the time spent on data cleaning, analysis, and visualization.

• **10.1.3** The project provides value by helping understand flight delays and supporting better planning and decision-making.

• **10.1.4** an interactive dashboard may be developed to present results more effectively, which may require additional effort.

11. Insights

• 11.1 at the end of the project, the analysis is expected to provide:

- Identify which airlines may have higher delay rates
- Understand the relationship between departure delay and arrival delay
- Discover whether certain days of the week have more delays
- Identify the most common causes of flight delays
- Detect general patterns and trends in flight performance
- Provide useful information that can support better planning and decision-making

12. Recommendations

• 12.1 Based on the expected analysis results, the project may recommend:

- Improving flight scheduling to reduce delays
- Focusing on airlines that show higher delay rates
- Improving planning during busy travel days
- Enhancing operations to reduce departure delays
- Preparing better strategies for weather-related disruptions
- Using data analysis regularly to support better decision-making